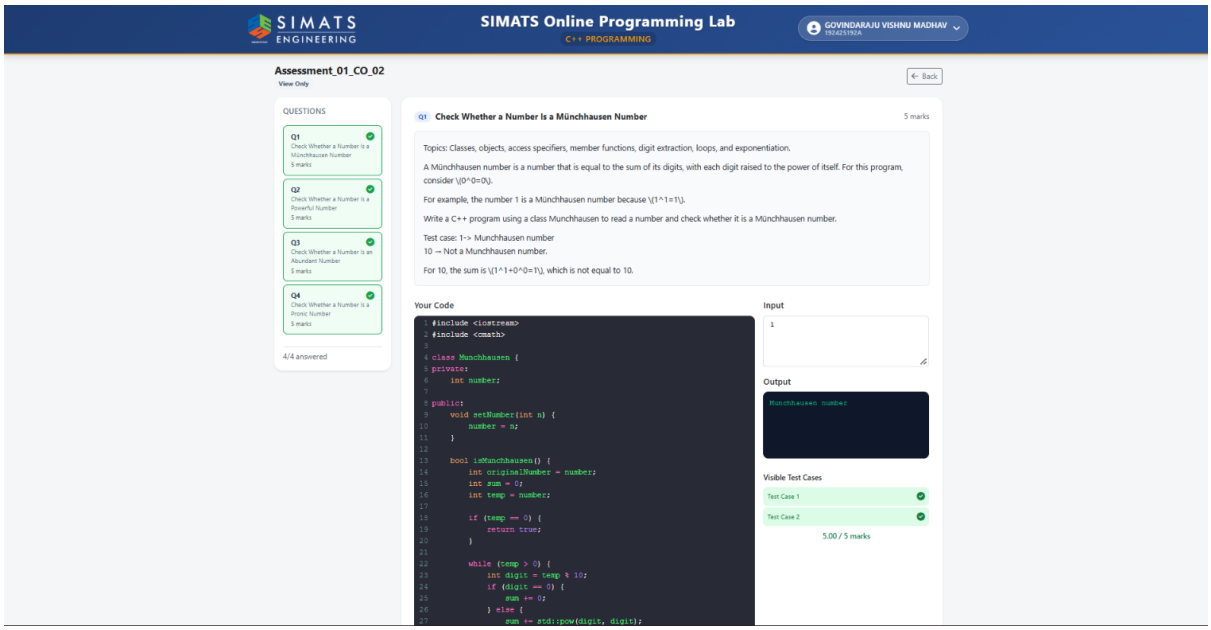
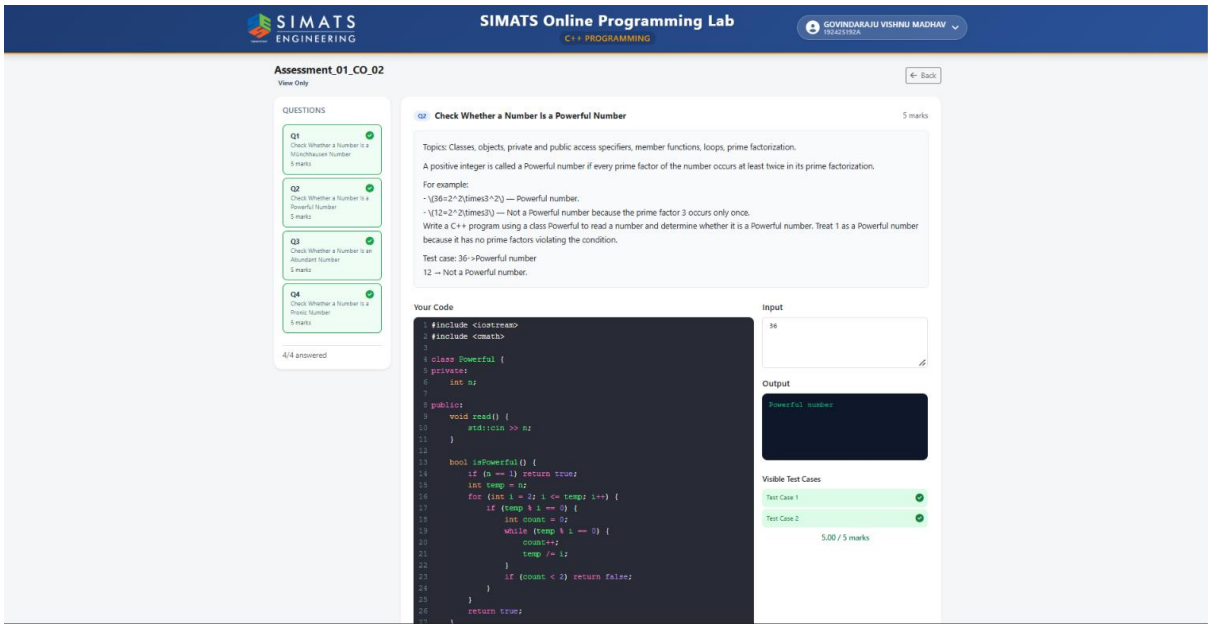
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C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assessment_01_CO_02
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Back
QUESTIONS
Q1 Check Whether a Number Is a Münchhausen Number 5 marks
Q2 Check Whether a Number Is a Powerful Number 5 marks
Q3 Check Whether a Number Is an Abundant Number 5 marks
Q4 Check Whether a Number Is a Pronic Number 5 marks
4/4 answered
Q1 Check Whether a Number Is a Münchhausen Number
5 marks
Topics: Classes, objects, access specifiers, member functions, digit extraction, loops, and exponentiation.
A Münchhausen number is a number that is equal to the sum of its digits, with each digit raised to the power of itself. For this program, consider \(0^0=0\).
For example, the number 1 is a Münchhausen number because \(1^1=1\).
Write a C++ program using a class Munchhausen to read a number and check whether it is a Münchhausen number.
Test case: 1-> Munchhausen number
10 → Not a Munchhausen number.
For 10, the sum is \(1^1+0^0=1\), which is not equal to 10.
Your Code
#include <iostream>
#include <cmath>
class Munchhausen {
private:
int number;
public:
void setNumber(int n) {
number = n;
}
bool isMunchhausen() {
int originalNumber = number;
int sum = 0;
int temp = number;
if (temp == 0) {
return true;
}
while (temp > 0) {
int digit = temp % 10;
if (digit == 0) {
sum += 0;
} else {
sum += std::pow(digit, digit);
Input
1
Output
Munchhausen number
Visible Test Cases
Test Case 1
Test Case 2
5.00 / 5 marks


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GOVINDARAJU VISHNU MADHAV
Assessment_01_CO_02
View Only
Back
QUESTIONS
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4/4 answered
Q2 Check Whether a Number Is a Powerful Number
5 marks
Topics: Classes, objects, private and public access specifiers, member functions, loops, prime factorization.
A positive integer is called a Powerful number if every prime factor of the number occurs at least twice in its prime factorization.
For example:
- \(36=2^2\times3^2\) — Powerful number.
- \(12=2^2\times3\) — Not a Powerful number because the prime factor 3 occurs only once.
Write a C++ program using a class Powerful to read a number and determine whether it is a Powerful number. Treat 1 as a Powerful number because it has no prime factors violating the condition.
Test case: 36->Powerful number
12 → Not a Powerful number.
Your Code
#include <iostream>
#include <cmath>
class Powerful {
private:
int n;
public:
void read() {
std::cin >> n;
}
bool isPowerful() {
if (n == 1) return true;
int temp = n;
for (int i = 2; i <= temp; i++) {
if (temp % i == 0) {
int count = 0;
while (temp % i == 0) {
count++;
temp /= i;
}
if (count < 2) return false;
}
}
return true;
Input
36
Output
Powerful number
Visible Test Cases
Test Case 1
Test Case 2
5.00 / 5 marks

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← Back

QUESTIONS

- Q1 Check Whether a Number Is a Munchhausen Number — 5 marks
- Q2 Check Whether a Number Is a Powerful Number — 5 marks
- Q3 Check Whether a Number Is an Abundant Number — 5 marks
- Q4 Check Whether a Number Is a Pronic Number — 5 marks

4/4 answered

Q3 Check Whether a Number Is an Abundant Number — 5 marks

Topics: Classes, objects, access specifiers, member functions, loops, divisors, arithmetic calculations.

A positive integer is called an Abundant number if the sum of its proper divisors is greater than the number itself.

Proper divisors are the positive divisors excluding the number itself.

For example, the proper divisors of 12 are 1, 2, 3, 4, and 6.

Their sum is:

\[
1+2+3+4+6=16
\]

Since \(16>12\), the number 12 is an Abundant number.

Write a C++ program using a class Abundant to read a number and check whether it is an Abundant number.

Test case: 12->Abundant number
15 → Not an Abundant number.

Your Code

```cpp
#include <iostream>

using namespace std;

class Abundant {
private:
    int number;

public:
    void readNumber() {
        cin >> number;
    }

    void checkAbundant() {
        int sum = 0;

        for (int i = 1; i <= number / 2; i++) {
            if (number % i == 0) {
                sum += i;
            }
        }
```

Input

```
12
```

Output

```
Abundant number
```

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓

5.00 / 5 marks

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Assessment_01_CO_02

View Only

← Back

QUESTIONS

- Q1 Check Whether a Number Is a Munchhausen Number — 5 marks
- Q2 Check Whether a Number Is a Powerful Number — 5 marks
- Q3 Check Whether a Number Is an Abundant Number — 5 marks
- Q4 Check Whether a Number Is a Pronic Number — 5 marks

4/4 answered

Q4 Check Whether a Number Is a Pronic Number — 5 marks

Topics: Classes, objects, private data members, public member functions, loops, multiplication, logical conditions.

A number is called a Pronic number if it can be expressed as the product of two consecutive integers.

For example:
- \(12=3\times4\) — Pronic number.
- \(20=4\times5\) — Pronic number.
- 15 cannot be expressed as the product of two consecutive non-negative integers, so it is not a Pronic number.

Write a C++ program using a class Pronic to read a number and check whether it is a Pronic number.

Test case: 20->Pronic number
15 → Not a Pronic number.

Your Code

```cpp
#include <iostream>
using namespace std;

class Pronic {
private:
    int num;
public:
    void input() {
        cin >> num;
    }
    void check() {
        bool isPronic = false;
        for (int i = 0; i * (i + 1) <= num; i++) {
            if (i * (i + 1) == num) {
                isPronic = true;
                break;
            }
        }
        if (isPronic) {
            cout << "Pronic number" << endl;
        } else {
            cout << "Not a Pronic number" << endl;
        }
    }
};

int main() {
```

Input

```
20
```

Output

```
Pronic number
```

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓

5.00 / 5 marks